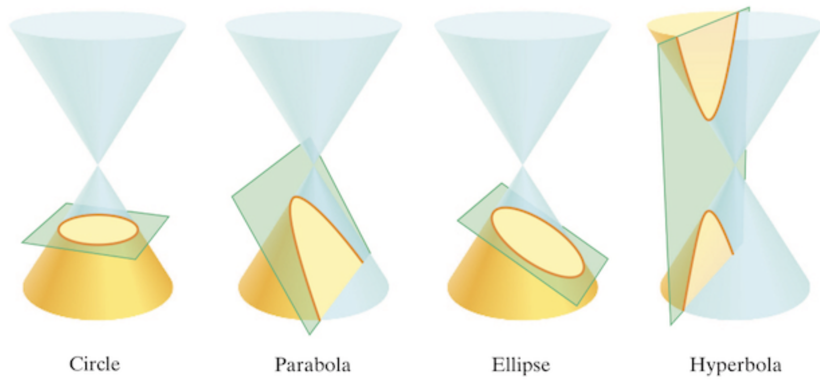
Ms. Welles **Alg 2 Trig/Honors**

Chp 11.1: Circles and Parabolas

Conic sections are named so because each conic section is the intersection of a right circular cone and a plane. The circle, parabola, ellipse, and hyperbola are the conic sections.



I. Circles

A circle is the set of all points in the plane that are equidistant from a fixed point C called the ___________________.

The equation of a circle centered at (h,k) with radius r is _______________________

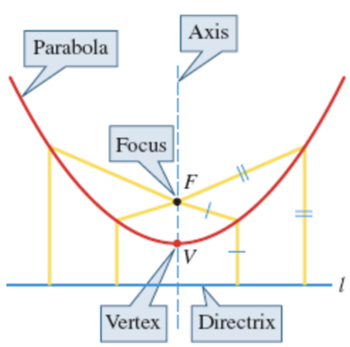
A circle with radius r centered at the origin has equation ___________________.

II. Parabolas

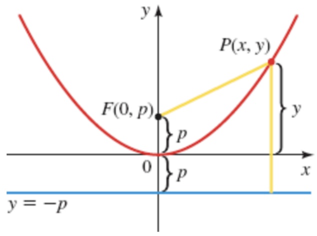
Recall from chapter 3 that the graph of the equation $y = ax^2 + bx + c$ is a __________________curve called a *parabola* that opens either upward or downward, depending on whether the number a is positive or negative.

Geometric Definition of a Parabola

A parabola is the set of all points in the plane that are equidistant from a fixed point F (called the **focus**) and a fixed line l (called the **directrix**). The **vertex** V of the parabola lies halfway between the focus and the directrix, and the **axis of symmetry** is the line that runs through the focus perpendicular to the directrix.



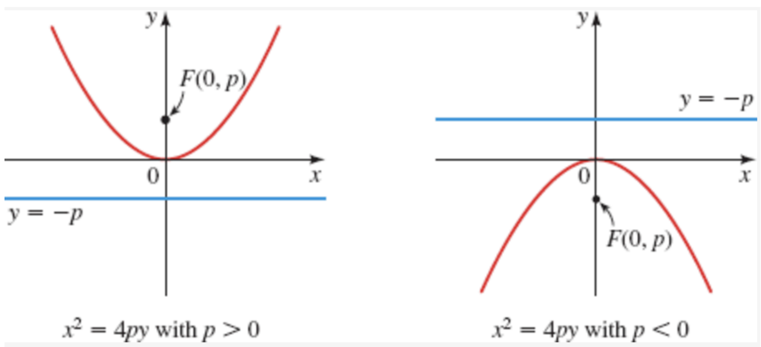
In this section we restrict our attention to parabolas that are situated with the **vertex at the origin** and that have a **vertical or horizontal axis of symmetry.**



Parabola with Vertical Axis

The standard equation of a parabola with vertical axis and vertex at the origin is $x^2 = 4py$. If $p > 0$, then the parabola opens upward; if $p < 0$, it opens downward. When x is replaced by $-x$, the equation remains unchanged, so the graph is symmetric about the ***y*-axis**.

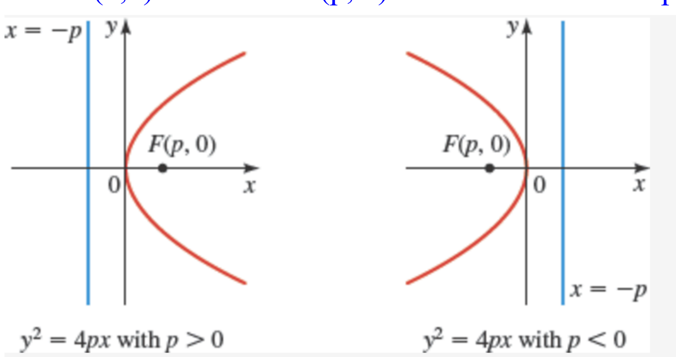
Vertex:(0,0) Focus: (0, p) Directrix: y = -p



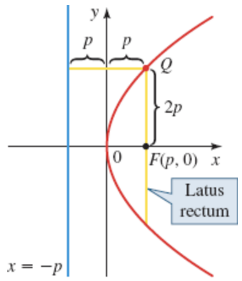
Parabola with Horizontal Axis

The standard equation of a parabola with horizontal axis and vertex at the origin is $y^2 = 4px$. If $p > 0$, then the parabola opens to the right; if $p < 0$, it opens to the left.

Vertex:(0,0) Focus: (p, 0) Directrix: x = - p



We can use the coordinates of the focus to estimate the “width” of a parabola when sketching its graph. The line segment that runs through the focus perpendicular to the axis, with endpoints on the parabola, is called the latus rectum, and its length is the focal diameter of the parabola.



Example 1: An equation of a parabola is given.

(a) Find the focus, directrix, and focal diameter of the parabola.

(b) Sketch a graph of the parabola and its directrix.

i. $x^2 = -8y$ ii. $y^2 = 24x$ iii. $5x + 3y^2 = 0$

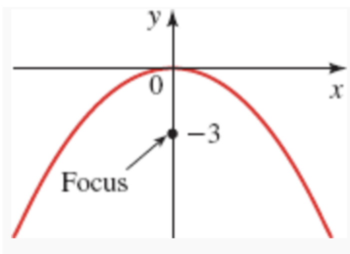
Example 2: Find the standard equation for the parabola that has its vertex at the origin and satisfies the given condition(s).

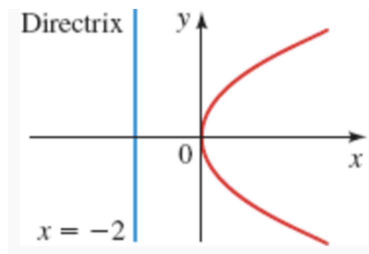
a. Focus: $\left(\frac{-1}{12}, 0\right)$ b. Directrix: $y = -5$

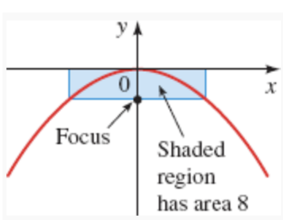
c. Focus on the negative y-axis, 6 units away from the directrix.

d. Opens upward with focus 5 units away from the vertex.

Example 3: Find the standard equation of the parabola whose graph is shown.

a. 

b. 

c. 

d.